Emergence Of Sodium-Ion Batteries

When scarcity of lithium required for lithium-ion batteries is affecting large-scale production of these batteries, the emergence of alternatives such as sodium-ion batteries is gaining importance. But will these be able to replace the lithium-ion batteries soon enough and prove better in the long run?

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In the battery dependent lifestyle that is the norm today, the coming of lithium-ion batteries was considered as a great breakthrough. Offering many advantages related to capacity, compactness, and life cycle, there is no doubt that lithium-ion-battery-chemistry reigns supreme in manifold electronic devices. However, of late, practical experiences with these batteries along with growing concerns over cost, safety (towards fires and explosions), and lithium production limitations are casting a doubt on the future of these batteries.

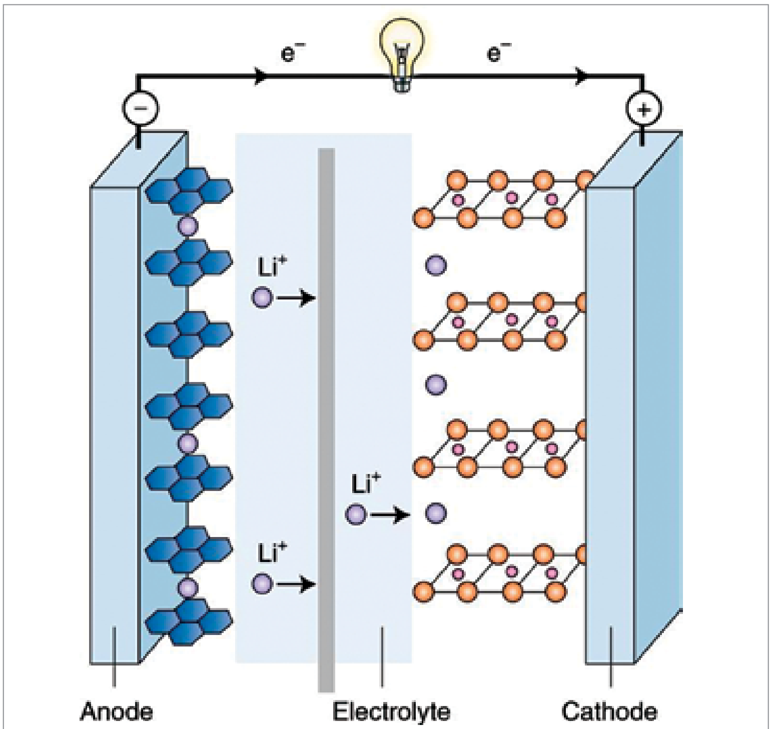

Fig. 1: Components of a rechargeable Li-ion battery (Credit: www.nature.com/articles)

Scientists, engineers, and technologists are constantly thinking about other advantageous battery options, and one possible alternative for lithium-ion (Li-ion) batteries is sodium-ion batteries. Recently, sodium-ion (Na-ion) batteries have emerged as another viable means of energy storage, mainly for large-scale electric storage applications. This is primarily due to low cost and easy availability of sodium as compared to lithium, while having similar chemistry and intercalation kinetics to that of lithium battery and the irreversible capacity of carbon anodes.

Recent research developments like those to increase sodium storage behaviour via electron-rich element-doped amorphous carbon are aiming to make Na-ion batteries a more affordable and sustainable replacement to Li-ion batteries. Therefore, Na-ion batteries are a recent development being promoted repeatedly not only as an economically promising alternative to lithium-ion batteries but also as one with high-performance and safe battery options.

Comparison with lithium-ion batteries

Conventional Li-ion batteries with liquid electrolyte can store high amounts of energy, whereas solid-state Na-ion batteries with a solid electrolyte core is unable to produce the same amounts of energy. Recent research however, has brought out a solid electrolyte that is as conductive as the liquid electrolytes used in lithium-ion batteries.

The last challenge towards achieving a highly efficient solid-state Na-ion battery was to find solid interfaces. Addressing this, the research has two main findings. The first